Testing Quantum Entanglement at the LHC with Electroweak Boson Pairs

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Abstract

In this report, we investigate quantum observables that quantify entanglement in bipartite systems of electroweak (EW) bosons, focusing on ZZ and W^+W^- production at the Large Hadron Collider (LHC) via proton-proton collisions. Using Monte Carlo simulations generated with `MadGraph5`, we reconstruct the spin density matrices of these processes from the angular distributions of their fully leptonic decay products. From these matrices, we compute a lower bound on the concurrence and evaluate the expectation value of the Bell operator to test for violations of the Bell inequality in dimension-3 spin systems (qutrits). This analysis provides a quantitative approach to identifying entanglement in High Energy Physics (HEP), using simulated data to test and validate the analytical predictions. We find that the ZZ system exhibits signs of entanglement, with the lower bound on the concurrence being mostly non-zero and reaching maximum values of $\mathcal{C}_{LB} \approx 0.7$, and the expectation value of the Bell operator reaching up to $\langle \mathcal{O}_B \rangle \lesssim 2.0$, with only a few regions of phase space violating the classical limit of 2. In contrast, the W^+W^- system shows slightly lower values for the concurrence bound, with $\mathcal{C}_{LB} \approx 0.6$, but displays more frequent instances where $\langle \mathcal{O}_B \rangle \gtrsim 2$, indicating a potentially stronger violation of the Bell inequality in specific kinematic regions.

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1 Introduction

Quantum entanglement lies at the heart of quantum mechanics, manifesting as correlations between spatially separated particles that defy classical expectations. Since the formulation of the Einstein-Podolsky-Rosen (EPR) paradox [1], which challenged the completeness of quantum mechanics by invoking local realism[1] and hidden variables, the nature of entanglement has remained a central topic in the foundations of physics. In particular, Bell's theorem [2] formalised these debates by demonstrating that no local hidden variable theory can reproduce all the predictions of quantum mechanics. This conclusion arises from the violation of Bell-type inequalities, most notably the CHSH inequality [3], which provides a testable constraint for local realistic theories.

Empirical violations of the CHSH inequality [4]—and later high-precision tests involving entangled photons [5, 6]—have strongly confirmed the nonlocal character of quantum correlations. These foundational results have inspired a broad spectrum of research into quantum entanglement, including its extension to systems of higher dimensionality through generalised inequalities such as the CGLMP inequality [7], which probe entanglement in multi-level systems.

The relevance of studying entanglement extends beyond the development of a complete quantum theory. It is also a foundational concept in quantum information theory [8], which is rapidly becoming a central pillar of modern physics. Quantum information theory has found applications in various fields, including quantum computing [9], quantum cryptography [10], and quantum machine learning [11]. As such, the study of entanglement is not merely a theoretical curiosity, but a practical necessity for the advancement of quantum technologies. In the quest for reliable entanglement measures, several quantifiers have been proposed, which will be discussed throughout this report.

In recent years, there has been growing interest in exploring quantum entanglement in the realm of high-energy physics [12], particularly through processes accessible at the Large Hadron Collider (LHC). Analyses from the **ATLAS** experiment [13, 14] have shown that the spin correlations of electroweak bosons—namely the $W^{\pm}$ and Z bosons—may encode signatures of entanglement, thus opening a new experimental frontier for probing quantum correlations in EW boson systems. Such studies are significant not only for deepening our understanding of the Standard Model [15], but also for advancing quantum information theory and the broader development of quantum technologies.

This report is structured as follows. In Section 2 we introduce the theoretical concepts from quantum information theory that quantify entanglement in our bipartite system of qutrits, as well as the definition of the coordinate system used to define the decay angles of the leptons that decay from our EW bosons. Also, we describe the different constructions involved in the process of reconstructing the density matrix. In Section 3 we describe the numerical setup for generating our events, as well as the phase space sampling method employed to produce graphs for comparison with the relevant literature. In Section 4 we present the results of our analysis, including an erratum to previous work conducted on calculating the angular coefficients of EW boson pairs [16], a discussion of the unphysicality of the density matrix, and the results of our analysis of the ZZ and W^+W^- production processes. Finally, in Section 5 we present our conclusions, in addition to an evaluation of the possible future directions of this work.

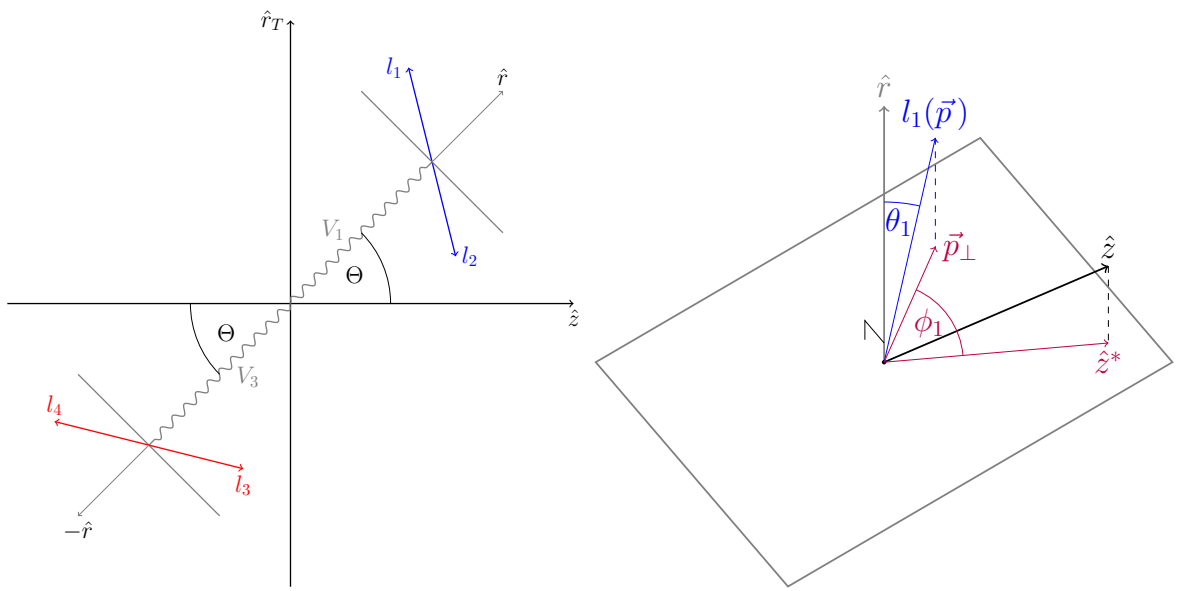
[1] Local realism is the combination of *locality*—information cannot travel faster than the speed of light—and *realism*—physical properties exist independently of measurement.

2 Theoretical Framework

In this section we introduce the concepts that form the theoretical foundations of the work we conducted. There are many different ways to signal the entanglement of a system, but in this work we will focus on two quantities: the lower bound on the concurrence $\mathcal{C}_{LB}$, and the expectation value of the Bell operator $\langle \mathcal{O}_B \rangle$, which we will use to test the violation of the Bell inequality. This report does not delve into the derivation of these quantities in full, but we will provide a brief overview of the mathematical formalism behind them, and how they can be calculated from the density matrix of our system. Both of these quantities require an accurate reconstruction of the density matrix of our system, for which we discuss two different methods of reconstruction. However, in order to reconstruct the density matrix, we need to define the coordinate system in which we will calculate the decay angles of the final state leptons.

2.1 Coordinate System

To reliably reconstruct the density matrix of the bipartite qutrit system, we must compute the decay angles of the lepton pairs that decay from the EW bosons. A key challenge lies in choosing a consistent coordinate system to define these angles, as the density matrix is highly sensitive to them. In this work, we adopt the modified helicity coordinate system, shown in Figure 1, originally introduced by the **ATLAS** collaboration [13].



(a) Diboson CM frame **(b)** Plane perpendicular to $\hat{r}$

Figure 1: Modified helicity coordinate system. **(a)** The individual boson rest frames are shown with different colours (red and blue). The scattering angle, Θ, is defined between the diboson flight path, $\hat{r}$, and the beam axis, $\hat{z}$. **(b)** Decay angles of a lepton. The azimuthal decay angle, ϕ_1, is measured between the lepton momentum and the beam axis $\hat{z}^*$, both projected in the perpendicular plane, while the polar angle θ_1 is measured between the lepton momentum and $\hat{r}$. Adapted from [17].

To extract the decay angles of the lepton pairs (l_1, l_2) and (l_3, l_4), their four-momenta are boosted

into the coordinate system shown in Figure 1. First, all leptons are boosted from the lab frame into the diboson center-of-mass (CM) frame using a boost vector derived from the sum of their four-momenta. Each lepton pair is then boosted into the rest frame of its parent boson, reconstructed from its decay products. This defines a common reference axis, the diboson flight path $\hat{r}$, allowing the decay angles to be measured consistently across different frames. It is important to note that the vector boson four-momenta are also boosted into the diboson CM frame, before the lepton pairs are boosted into the rest frame of their parent boson. The formalism for this was described in greater detail in previous works [17], and will not be repeated here, for brevity.

2.2 Introduction to the Density Matrix

The density matrix is a mathematical tool used to describe quantum systems that are in a statistical mixture of pure states $|\psi_i\rangle$, each occurring with probability p_i, within the Hilbert space $\mathcal{H}$ of the system. Hence, a density matrix can describe both pure and mixed states, but what do we mean by mixed states? When we talk about mixed and pure states, we are referring to classical probabilities of the system being in a certain state. Although a pure state may be a superposition, if the state vector is exactly known, it is considered pure. Hence, we define a mixed state as a linear combination of pure states, each weighted by the classical probability associated with that pure state. Mathematically, it is easiest to describe pure and mixed states using the density matrix formalism,

$$\rho = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right| \ . \tag{2.2.1}$$

This formalism is useful because it allows us to compute the expectation value of some quantum operator, A, by using the following relation [18],

$$\langle A\rangle = \mathrm{Tr}\{\rho A\} \, . \tag{2.2.2}$$

In addition, for a density matrix to represent a valid quantum state, it must satisfy the following three conditions [9]:

1. **Hermiticity:** The density matrix must be Hermitian.
2. **Trace:** The trace of the density matrix must be normalised such that it equals 1.
3. **Positivity:** The density matrix must be positive semi-definite (PSD), i.e., all eigenvalues of ρ must be non-negative.

2.3 Quantum Observables

We have introduced already the density matrix, which is a mathematical object that encodes the information of a quantum system. Here we will discuss how we can use the density matrix to calculate quantum observables which could indicate the presence of entanglement in our system.

2.3.1 Purity and Concurrence

The concept of purity is a measure of how mixed our density matrix is. For example, consider a density matrix ρ which is a pure state,

$$\rho = \left|\psi_1\right\rangle \left\langle\psi_1\right| \ , \tag{2.3.1}$$

where $|\psi_1\rangle$ is some basis state, then we can define the purity, $p(\rho)$, of the density matrix as

$$p(\rho) = \mathrm{Tr}\{\rho^2\} = \mathrm{Tr}\{|\psi_1\rangle\langle\psi_1||\psi_1\rangle\langle\psi_1|\} = \mathrm{Tr}\{|\psi_1\rangle\langle\psi_1|\} = \mathrm{Tr}\{\rho\} = 1\,, \tag{2.3.2}$$

therefore we see that a pure state has a purity of 1. Whereas, if we had a mixed state, for example,

$$\rho = \frac{3}{8}|\psi_1\rangle\langle\psi_1| + \frac{5}{8}|\psi_2\rangle\langle\psi_2|\ , \tag{2.3.3}$$

we would have

$$p(\rho) = \mathrm{Tr}\Big\{\frac{9}{64}|\psi_1\rangle\langle\psi_1| + \frac{25}{64}|\psi_2\rangle\langle\psi_2|\Big\} = \frac{34}{64} < 1\,, \tag{2.3.4}$$

so the purity of a mixed state is less than 1.

Now that we have defined the purity of a density matrix, this leads us to the definition of concurrence, first introduced in [19]. More precisely, we will be using the lower bound on the concurrence, $\mathcal{C}_{LB}$, since the actual concurrence of a system can only be derived analytically for two qubits [20]. The concurrence of a pure state in arbitrary dimensions has been discussed in [21], but since we are interested in mixed states, we will use the lower bound on the concurrence [22, 23]. This can still be used to quantify the entanglement present in a density matrix of a bipartite system, since a non-zero lower bound indicates entanglement. However, before we do, there is one more piece of technology in terms of quantum information theory that we need to introduce, which is the partial trace [24]. The partial trace is a mathematical operation that allows us to obtain the reduced density matrix of a subsystem of a larger system. For example, if we have a density matrix ρ, which represents a bipartite system in the Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, we can obtain the reduced density matrix of subsystem A by taking a partial trace over ρ with respect to the Hilbert space $\mathcal{H}_B$,

$$\rho_A = Tr_B(\rho) = \sum_b (\hat{\mathbf{1}} \otimes \langle\hat{b}|)\rho(\hat{\mathbf{1}} \otimes |\hat{b}\rangle)\,, \tag{2.3.5}$$

where $|\hat{b}\rangle$ is the orthonormal basis of the Hilbert space $\mathcal{H}_B$, and $\hat{\mathbf{1}}$ is the identity operator in the Hilbert space $\mathcal{H}_A$. The lower bound on the concurrence is defined as [22, 23],

$$\mathcal{C}_{LB}^2 = 2\max\left(0,\, \mathrm{Tr}\{\rho^2\} - \mathrm{Tr}\{\rho_A^2\},\, \mathrm{Tr}\{\rho^2\} - \mathrm{Tr}\{\rho_B^2\}\right), \tag{2.3.6}$$

where ρ_A and ρ_B are the reduced density matrices of the two subsystems, in our case the two qutrits. We can see that the lower bound of the concurrence is the difference between the purity of the full density matrix and the purity of the reduced density matrices.

2.3.2 Bell Inequalities

We continue our discussion of entanglement by introducing Bell-like inequalities, which serve as the primary indicators of quantum entanglement. Bell inequalities are constraints that must be satisfied by any theory based on local realism [2], and their experimental violation [4, 5] is strong evidence of quantum entanglement. For two qubits (spin-1/2), the standard test is the CHSH inequality [3]. However, since we are studying qutrits (spin-1), a generalisation to higher dimensions is needed. The CGLMP inequality [7] is suitable for qutrit systems and involves two measurement operators per subsystem: $\hat{A}_1$, $\hat{A}_2$ acting on $\mathcal{H}_A$ and $\hat{B}_1$, $\hat{B}_2$ on $\mathcal{H}_B$, each with three possible outcomes corresponding to spin projections. The CGLMP inequality is given by:

$$\begin{aligned}\mathcal{I}_3 =& P(A_1 = B_1) + P(B_1 = A_2 + 1) + P(A_2 = B_2) + P(B_2 = A_1) \\ &- P(A_1 = B_1 - 1) - P(B_1 = A_2) - P(A_2 = B_2 - 1) - P(B_2 = A_1 - 1) \leq 2,\end{aligned} \tag{2.3.7}$$

where $P(A_i = B_j + k)$ denotes the probability that the measurement outcomes differ by k modulo d, with $d = 3$ in our case [7]. These probabilities can be expressed as expectation values using the density matrix and projection operators, for example,

$$P(B_1 = A_2 + 1) = \langle \mathcal{P}^x_{B_1} \otimes \mathcal{P}^y_{A_2} \rangle = \mathrm{Tr}\{\rho\, \mathcal{P}^x_{B_1} \otimes \mathcal{P}^y_{A_2}\}, \tag{2.3.8}$$

where $\mathcal{P}^x_{B_1}$ is the projection operator corresponding to the measurement of B_1 giving a value x, and $\mathcal{P}^y_{A_2}$ is the projection operator corresponding to the measurement of A_2 giving a value y. Here, $x, y \in \{1, 0, -1\}$, and since $k = 1$, we require $|x-y| = 1 \bmod 3$. This condition is satisfied by four pairs of values: $(x, y) = (1, 0), (0, -1), (-1, 0), (0, 1)$. Therefore, four pairs of projection operators contribute to this probability, each corresponding to one of the valid combinations of x and y that satisfy the modular condition. Hence, the entire inequality can be written compactly as the expectation value of a 9-dimensional square matrix referred to as the Bell operator, $\mathcal{O}_B$, which encodes all of the projection operators into one:

$$\mathcal{I}_3 = \langle \mathcal{O}_B \rangle = \mathrm{Tr}\{\rho\, \mathcal{O}_B\} \leq 2, \tag{2.3.9}$$

We can maximise the violation of the Bell inequality by optimising the choice of measurement basis for the Bell operator. The inequality in Equation 2.3.7 is constructed in terms of general measurement operators, and by applying a change of basis, we obtain a different realisation of these operators. This change of basis can be implemented by applying a unitary transformation to the Bell operator. Specifically, we introduce two unitary matrices $U, V \in U(3)$ that act on the Hilbert spaces of the two subsystems. The parameter space over which we optimise the Bell operator is given by the set of all possible unitary transformations, hence by parameterising the unitary matrices and varying U and V, we can find the maximum violation of the Bell operator. This can be interpreted as probing different measurement settings for the two subsystems, effectively changing the measurement operators A_1, A_2, B_1, B_2. Optimising the inequality shown in Equation 2.3.7, we can find the maximum violation of the Bell operator [25, 26, 27],

$$\mathcal{I}_3 = \langle \mathcal{O}_B \rangle_{max} = Tr\{\rho (U \otimes V)^\dagger \mathcal{O}'_B (U \otimes V)\}\,. \tag{2.3.10}$$

The maximal violation of the Bell operator is given by the transformation of $\mathcal{O}'_B$ which can be expressed as [26],

$$\mathcal{O}'_B = -\frac{2}{\sqrt{3}} \Big(\mathcal{S}_x \otimes \mathcal{S}_x + \mathcal{S}_y \otimes \mathcal{S}_y \Big) + \lambda_4 \otimes \lambda_4 + \lambda_5 \otimes \lambda_5\,. \tag{2.3.11}$$

This is the quantum operator associated with the CGLMP inequality. The matrices $\mathcal{S}_x$ and $\mathcal{S}_y$ are the spin-1 operators in the x and y directions, respectively, and λ_4 and λ_5 are the Gell-Mann matrices corresponding to the 4th and 5th generators of SU(3) (See Appendix A).

2.4 Mathematical Structure of the Density Matrix

Before we can calculate any quantum observable in the pursuit of quantifying the entanglement of our system, we need to accurately reconstruct the density matrix. To do this, we must choose a parametrisation of the density matrix, and then we can use the angular distributions (obtained from our data) of the leptons to reconstruct the density matrix. We do this by following the formalism laid out in [28], and choosing appropriate projector functions (not to be confused with the projectors defined in Section 2.3.2) that invert the angular distributions of the leptons into the coefficients of the density matrix. By 'invert' we mean that typically the angular distributions of the leptons can be derived analytically, by using the density matrix to find the differential cross-section, $d\sigma/d\Omega$, of the process [27, 29]. However, we aim to invert this procedure and reconstruct the density matrix from

the angular distributions, which are available from our MC simulations. Furthermore, we present two different parametrisations of the density matrix, one for the ZZ system in Section 2.4.1 and one for the W^+W^- system in Section 2.4.3.

2.4.1 Irreducible Tensor Operator Parametrisation

Below, we summarise the irreducible tensor operator (ITO) parametrisation of the ZZ density matrix, ρ [30, 31]. It expresses ρ as a linear combination of rank-l tensor operators T^l_m (See Appendix B), for each boson individually and also as a tensor product representing the correlation terms,

$$\rho = \frac{1}{9}\Big[\,\mathbf{1}_3 \otimes \mathbf{1}_3 + \sum A^{(1)}_{l,m} T^l_m \otimes \mathbf{1}_3 + \sum A^{(3)}_{l,m} \mathbf{1}_3 \otimes T^l_m + \sum C_{l_1,m_1,l_3,m_3} T^{l_1}_{m_1} \otimes T^{l_3}_{m_3}\Big]\,. \tag{2.4.1}$$

Where $\mathbf{1}_3$ is the 3-dimensional identity matrix, l is summed over 1 and 2 and m is summed over the range $-l$ to l, and similarly for l_1, m_1 and l_3, m_3. To calculate the coefficients A and C, we first relate them to the coefficients α and γ, used in [32], by the following relations,

$$A^{(k)}_{1,m} = -\frac{\sqrt{8\pi}}{\eta}\,\alpha^{(k)}_{1,m}, \qquad A^{(k)}_{2,m} = \sqrt{40\pi}\,\alpha^{(k)}_{2,m}, \tag{2.4.2}$$

$$C_{1,m_1,1,m_3} = \frac{8\pi\,\gamma_{1,m_1,1,m_3}}{\eta^2}, \qquad C_{2,m_1,2,m_3} = 40\pi\,\gamma_{2,m_1,2,m_3}, \tag{2.4.3}$$

$$C_{1,m_1,2,m_3} = -\frac{8\pi\sqrt{5}\,\gamma_{1,m_1,2,m_3}}{\eta}, \qquad C_{2,m_1,1,m_3} = -\frac{8\pi\sqrt{5}\,\gamma_{2,m_1,1,m_3}}{\eta}\,. \tag{2.4.4}$$

Where parameter η is defined as the relative balance of the left and right chirality couplings of the vector bosons to massless leptons, with $\eta^{(W)} = 1$ and $\eta^{(Z)} = 0.214$. We write the coefficients in this way due to the ease of calculating the α and γ coefficients, as briefly discussed in the next section. The focus of previous work [17], was to reliably calculate these coefficients, and compare our results to the literature, namely [32].

2.4.2 Relation to Spherical Harmonics

The α and γ coefficients shown in Equation 2.4.2 are related to the angular distributions of the leptons and can be calculated by using the spherical harmonic functions [33], as discussed in previous work [17]. Here, we will briefly show how to calculate these coefficients to use them in the density matrix. The angular distributions of the leptons can be written analytically as a linear combination of spherical harmonics $Y_{l,m}(\theta,\phi)$, with the α and γ coefficients appearing as the corresponding weights. By exploiting the orthonormality of the spherical harmonics, these coefficients can be extracted directly from the distributions via projection integrals. Therefore, we can write the coefficients as expectation values of the spherical harmonics,

$$\begin{aligned} \alpha^{(k)}_{l,m} &= \langle Y_{l,m}(\theta_k,\phi_k)\rangle\,, \\ \gamma_{l_1,m_1,l_3,m_3} &= \langle Y_{l_1,m_1}(\theta_1,\phi_1)\,Y_{l_3,m_3}(\theta_3,\phi_3)\rangle\,. \end{aligned} \tag{2.4.5}$$

Where the angles θ_k and ϕ_k are the polar and azimuthal angles of the leptons in the modified helicity coordinate system discussed in Section 2.1.

This gives us a complete method to reconstruct the density matrix for a bipartite system of vector bosons, and is particularly useful in modelling the system of two Z bosons, since the chiral coupling

of the Z boson can easily be modeled through the parameter η. Therefore, we will be using this parameterisation for the density matrix of the ZZ boson system. In addition, another important thing to note is that since the Z bosons are indistinguishable, we must symmetrise the density matrix, which means that we must average the coefficients under the exchange of indices,

$$\begin{aligned} A^{(1)}_{l,m} &= A^{(3)}_{l,m} = \frac{1}{2}\Big(A^{(1)}_{l,m} + A^{(3)}_{l,m}\Big)\,, \\ C_{l_1,m_1,l_3,m_3} &= \frac{1}{2}\Big(C_{l_1,m_1,l_3,m_3} + C_{l_3,m_3,l_1,m_1}\Big)\,. \end{aligned} \tag{2.4.6}$$

2.4.3 Gell-Mann Basis

Alternatively, one can expand ρ in the basis of Gell-Mann matrices, which is often used for the W^+W^- system [25, 26, 28],

$$\rho = \frac{1}{9}\,\mathbf{1}_3 \otimes \mathbf{1}_3 + \frac{1}{3}\sum_{i=1}^{8} f_i\,\lambda_i \otimes \mathbf{1}_3 + \frac{1}{3}\sum_{j=1}^{8} g_j\,\mathbf{1}_3 \otimes \lambda_j + \sum_{i=1}^{8}\sum_{j=1}^{8} h_{ij}\,\lambda_i \otimes \lambda_j\,, \tag{2.4.7}$$

where λ_i are the orthonormal, traceless Gell-Mann matrices [34], and f_i, g_j and h_{ij} are coefficients that can be calculated from the angular distributions of the decay products of the vector bosons. The Gell-Mann matrices are normalised such that,

$$\mathrm{Tr}\{\lambda_i \lambda_j\} = 2\,\delta_{ij}\,, \tag{2.4.8}$$

so the coefficients in Equation 2.4.7 can be calculated by using this orthogonality principle,

$$\begin{aligned} f_i &= \frac{1}{2}\,\mathrm{Tr}\{\rho\,(\lambda_i \otimes \mathbf{1}_3)\} = \frac{1}{2}\,\langle \lambda_i \otimes \mathbf{1}_3 \rangle\,, \\ g_j &= \frac{1}{2}\,\mathrm{Tr}\{\rho\,(\mathbf{1}_3 \otimes \lambda_j)\} = \frac{1}{2}\,\langle \mathbf{1}_3 \otimes \lambda_j \rangle\,, \\ h_{ij} &= \frac{1}{4}\,\mathrm{Tr}\{\rho\,(\lambda_i \otimes \lambda_j)\} = \frac{1}{4}\,\langle \lambda_i \otimes \lambda_j \rangle\,. \end{aligned} \tag{2.4.9}$$

As shown in [28], these expectation values correspond to the expectation values of the projectors $\pi_i^{\pm}$, which are defined as [25],

$$\begin{aligned} \pi_1^{\pm} &= \sqrt{2}\sin\theta^{\pm}\cos\phi^{\pm}(5\cos\theta^{\pm} \pm 1), & \pi_2^{\pm} &= \sqrt{2}\sin\theta^{\pm}\sin\phi^{\pm}(5\cos\theta^{\pm} \pm 1), \\ \pi_3^{\pm} &= \frac{1}{4}(5 \pm 4\cos\theta^{\pm} + 15\cos(2\theta^{\pm})), & \pi_4^{\pm} &= 5\sin^2\theta^{\pm}\cos(2\phi^{\pm}), \\ \pi_5^{\pm} &= 5\sin^2\theta^{\pm}\sin(2\phi^{\pm}), & \pi_6^{\pm} &= \sqrt{2}\sin\theta^{\pm}\cos\phi^{\pm}(-5\cos\theta^{\pm} \pm 1), \\ \pi_7^{\pm} &= \sqrt{2}\sin\theta^{\pm}\sin\phi^{\pm}(-5\cos\theta^{\pm} \pm 1), & \pi_8^{\pm} &= \frac{1}{4\sqrt{3}}(-5 \pm 12\cos\theta^{\pm} - 15\cos(2\theta^{\pm})). \end{aligned} \tag{2.4.10}$$

Where the $\pm$ refers to the sign of the parent boson. We can use these functions to perform the inverse mapping from the angular distributions of the leptons to the spin operators shown in the density

matrix. Therefore, the coefficients f_i, g_j and h_{ij} can be calculated by using the following relations,

$$\begin{aligned} f_i &= \frac{1}{2}\langle \pi_i^+ \rangle\,, \\ g_j &= \frac{1}{2}\langle \pi_j^- \rangle\,, \\ h_{ij} &= \frac{1}{4}\langle \pi_i^+ \pi_j^- \rangle\,. \end{aligned} \tag{2.4.11}$$

This provides an alternative approach for reconstructing the density matrix, which is particularly useful for the W boson system, since the functions in Equation 2.4.10 are the projectors needed to reconstruct the density matrix. However, for the Z boson system, this is not the case, and the formalism must be extended to take into account the different chiral couplings of the Z boson to the leptons. The general method to do this is discussed in detail in [28], however we will use the ITO parameterisation as previously mentioned, since it is more convenient for our analysis. We also use this parameterisation for the W^+W^- system as a cross-check, but we do not show the results for this analysis as it is very similar to that already shown in Section 4.

3 Methodology

In order to reconstruct the density matrix, we need to generate the angular distributions of the decay products of the vector bosons. We study two processes: W^+W^- and ZZ production, at the LHC, with CM energy of 13 TeV. The events were generated at leading order (LO), and the leptonic decay channels were investigated, to be able to probe the spin structure of the system in a clean and precise manner. The leptonic decay channels we study are:

$$\begin{aligned} p\,p &\to e^+\,e^-\,\mu^+\,\mu^- \;\; (ZZ), \\ p\,p &\to e^+\,\nu_e\,\mu^-\,\bar{\nu}_\mu \;\; (W^+W^-)\,. \end{aligned}$$

3.1 Numerical setup

In this section, we outline the numerical setup used to generate the events for the W^+W^- and ZZ systems. The setup is identical to that used in previous work [17], and has been included here for completeness. All event simulations were performed using `MadGraph5` [35], within the five-flavour scheme. This framework treats all fermions as effectively massless relative to the electroweak bosons, except for the top quark. The analysis employed a *fully inclusive* event selection, based on the same input parameters as those outlined in [36]. Specifically, an invariant mass window was applied to same-flavour, oppositely charged lepton pairs:

$$81\,\text{GeV} < \mathrm{M}_{l^+l^-} < 101\,\text{GeV}. \tag{3.1.1}$$

The pole masses of the W and Z bosons, denoted M_V, were derived from their on-shell values M_V^{ON} [15]:

$$\begin{aligned} M_Z^{\mathrm{ON}} &= 91.1876\,\text{GeV}, \qquad & M_W^{\mathrm{ON}} &= 80.379\,\text{GeV}, \\ \Gamma_Z^{\mathrm{ON}} &= 2.4952\,\text{GeV}, \qquad & \Gamma_W^{\mathrm{ON}} &= 2.085\,\text{GeV}, \end{aligned} \tag{3.1.2}$$

using the standard conversion formulas [37]:

$$\mathrm{M}_V = \frac{M_V^{\mathrm{ON}}}{\sqrt{1+\left(\Gamma_V^{\mathrm{ON}}/M_V^{\mathrm{ON}}\right)^2}}, \qquad \Gamma_V = \frac{\Gamma_V^{\mathrm{ON}}}{\sqrt{1+\left(\Gamma_V^{\mathrm{ON}}/M_V^{\mathrm{ON}}\right)^2}}. \tag{3.1.3}$$

Parton distribution functions (PDFs) and the QCD coupling α_s were sourced from `LHAPDF6` [38], specifically using the `NNPDF30_nnlo_as_0118` set [39], where α_s is fixed at 0.118. The factorisation and renormalisation scales were both defined as the average of the masses of the two vector bosons involved in the event, i.e., $\mu_F = \mu_R = (\mathrm{M}_{V_1} + \mathrm{M}_{V_2})/2$.

3.2 Phase Space Sampling

The primary focus of this report is the study of the quantum observables of each system. In order to be able to better interpret the results of the quantum observables, we want to see how the observables behave in different regions of the phase space. By phase space, we mean the kinematic variables of the diboson system, not the decay angles of the leptons. Our phase space is defined by two variables: the invariant mass of the diboson system, M_{VV}, and the scattering angle of the diboson system, Θ, as discussed in Section 2.1. Therefore, we want to be able to compute the quantum observables for each region of this phase space, similar to the work done in [25, 26]. That way, we can see if the entanglement is only present in a restricted region of the phase space, so that future studies can focus on this region. To do this, however, is challenging from a computational point of view, since we need to be able to sample the phase space in a way that doesn't introduce any bias into the results. This can be done by utilising a custom phase space sampling, by hard coding cuts into the event generation in `MadGraph5`. We generate the same amount of events for each region and our hard coded cuts are as follows:

$$\begin{aligned} i \leq \mathrm{M}_{VV} &\leq i + 50\,\mathrm{GeV} \\ j \leq \cos\Theta &\leq j + 0.1\,, \end{aligned}$$

for $i \in \{200, 250, ..., 950\}$ for M_{ZZ} and $i \in \{200, 250, ..., 1150\}$ for M_{WW}, and $j \in \{0, 0.1, ..., 0.8\}$ for $\cos\Theta$. The reason why the invariant mass limits for the two systems differ is that we aim to study the W^+W^- system in greater depth, as it is the main focus of this work. Furthermore, the scattering angle limits are truncated before $\cos\Theta = 1$, since the events in this region are much more saturated towards the boundary of the phase space, therefore we decided to omit these regions from our analysis. In addition, it should be noted that the range of $\cos\Theta$ is from -1 to 1, however we only sample the positive half of the phase space, since the system is symmetric around the $\cos\Theta = 0$ axis, as can be seen in [26].

3.3 Eigenvalue Clipping

Once we have reconstructed the density matrix, we need to check if it is a valid density matrix. This means that the eigenvalues of the density matrix must be real and positive, as per the conditions set out in Section 2.2. In the presence of any negative eigenvalues, we can use a method referred to as a Higham projection [40], to project our density matrix onto the nearest valid density matrix, i.e., the nearest positive semi-definite matrix. This is done by setting any negative eigenvalues of the density matrix to zero, and then projecting the diagonal matrix of clipped eigenvalues back onto the original

vector space using the eigenvectors of the original density matrix. This procedure can be visualised mathematically as follows,

$$\rho' = M \cdot \mathrm{diag}(\mu_i') \cdot M^\dagger \,, \tag{3.3.1}$$

where M is the matrix of eigenvectors of the original density matrix, μ_i' are the clipped eigenvalues, and ρ' is the new density matrix.

3.4 Error Propagation

In this section, we will discuss the error propagation of the Bell operator, and how the covariance of the coefficients in the density matrix was taken into account. We also present an estimation on the uncertainty of the eigenvalues of the density matrix, which is important for the validity analysis of the density matrix. In general, the variance of some function $f(x_1, x_2, \ldots, x_n)$ can be calculated using the following relation [41],

$$\sigma_f^2 = \sum_{a,b=1}^{n} \left(\frac{\partial f}{\partial x_a}\right)\left(\frac{\partial f}{\partial x_b}\right)\sigma_{ab}^2 \,, \tag{3.4.1}$$

where σ_{ab}^2 is the covariance between the variables x_a and x_b. When $a = b$, this is simply the variance of the variable x_a. For the Bell operator, we can write its variance, $\sigma_{\mathcal{O}_B}^2$, as a function of the coefficients in the density matrix. By taking the parameterisation shown in Equation 2.4.7 as an example, we can expand out the expression in Equation 2.3.9,

$$\begin{aligned}\langle \mathcal{O}_B \rangle = &\frac{1}{9}\,\mathrm{Tr}\{\mathbf{1}_3 \otimes \mathbf{1}_3 \cdot \mathcal{O}_B\} + \frac{1}{3}\sum_{i=1}^{8} f_i\,\mathrm{Tr}\{\lambda_i \otimes \mathbf{1}_3 \cdot \mathcal{O}_B\} \\ &+ \frac{1}{3}\sum_{j=1}^{8} g_j\,\mathrm{Tr}\{\mathbf{1}_3 \otimes \lambda_j \cdot \mathcal{O}_B\} + \sum_{i,j=1}^{8} h_{ij}\,\mathrm{Tr}\{\lambda_i \otimes \lambda_j \cdot \mathcal{O}_B\}\,.\end{aligned} \tag{3.4.2}$$

We can see that the partial derivatives of the Bell operator with respect to the coefficients f_i, g_j and h_{ij}, that appear in the variance from Equation 3.4.1, will simply be trace terms. This means that we can compute the variance of the Bell operator using Equation 3.4.1, with $x_a, x_b \in \{f_i, g_j, h_{ij}\}$, and the covariance σ_{fgh}^2 being the covariance between the coefficients in the density matrix. In practice, this can be done by calculating the full covariance matrix, using the covariance function from `numpy` [42] for all 80 coefficients, and then constructing the vector representing the partial derivative terms. Mathematically, this can be expressed as,

$$\sigma_{\mathcal{O}_B}^2 = D^\dagger \cdot \sigma_{fgh}^2 \cdot D \,, \tag{3.4.3}$$

where D is the vector of partial derivatives of the Bell operator with respect to the coefficients in the density matrix, and σ_{fgh}^2 is the covariance matrix of the coefficients in the density matrix.

In order to estimate the uncertainty of the eigenvalues of the density matrix, the process is similar but slightly less straightforward. We still need to calculate the covariance matrix of the coefficients in the density matrix, but for a different purpose. We can find the standard deviation of the eigenvalues via Monte Carlo sampling [43], by generating different density matrices using a multivariate Gaussian distribution (again using the `numpy` function - which takes in the covariance matrix as an argument) for the coefficients in the density matrix. Then, we calculate the eigenvalues of each density matrix, and from this distribution we can calculate the standard deviation of the eigenvalues.

4 Results

In this section, we present the results of our analysis, including an erratum to the angular coefficients calculated in [17]. We then briefly discuss the validity of our density matrices by introducing a quantity called the unphysicality. Finally, we discuss the results for each bipartite system of EW bosons, W^+W^- and ZZ, presenting the concurrence and Bell operator results for each system, as well as the phase space checks. It should be noted that all results for the concurrence and Bell operator plots have been calculated with density matrices that have already been projected to be PSD, as discussed in Section 3.3.

4.1 Angular coefficients

In previous work [17], the angular coefficients from Equation 2.4.5 were calculated for the *fully-inclusive* ZZ setup at leading order (LO), and compared to Table 3 in [32]. However, a notable discrepancy was discovered in the coefficients with azimuthal dependence ($m \neq 0$). After contacting the authors of the study, it was confirmed that the discrepancy was due to a difference in the normalisation of the spherical harmonic functions used to calculate these coefficients. The authors of [32] normalised their spherical harmonics such that the real part was normalised to 1,

$$\int [\mathrm{Re}\{Y_{l,m}(\theta,\phi)\}]^2 \sin\theta d\theta d\phi = 1\,, \tag{4.1.1}$$

whereas we normalised the full complex functions to be 1,

$$\int [Y_{l,m}(\theta,\phi)]^* Y_{l,m}(\theta,\phi) \sin\theta d\theta d\phi = 1\,, \tag{4.1.2}$$

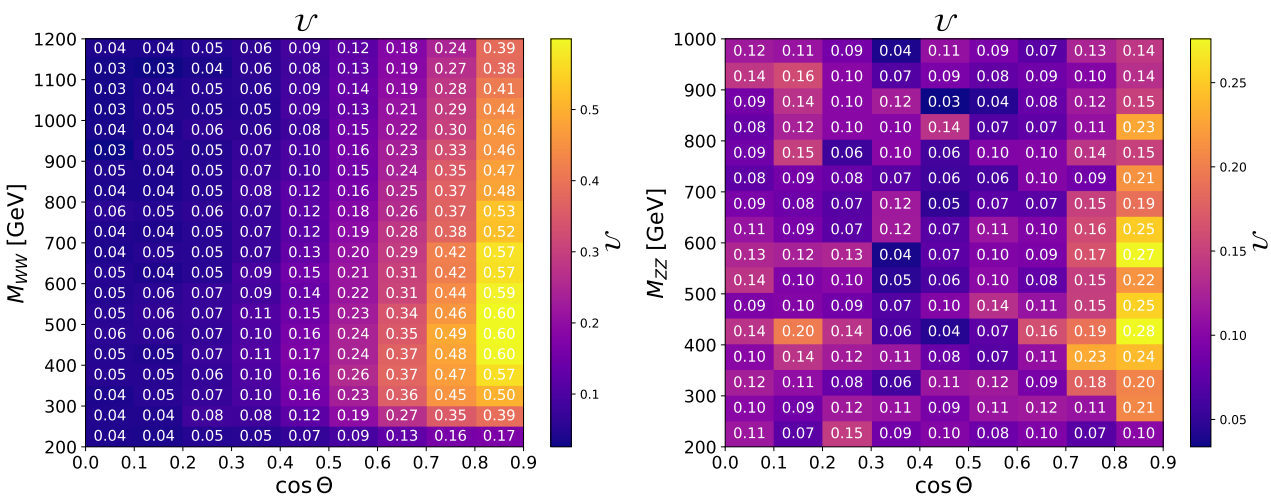
and then took the real part of the expectation value of the spherical harmonic when comparing our results. This leads to a missing factor of $\sqrt{2}$ in our definition, compared to the literature. Furthermore, a factor of $\sqrt{3}$ was also missing in the definition of the spherical harmonics in [32]. Taking these discrepancies into account, our updated results can be seen in Table 1, and are in very good agreement with the results in [32], once the factor of $\sqrt{3}$ is incorporated; giving values for $\alpha^{(1)}_{2,-2}$ and $\alpha^{(3)}_{2,-2}$ of -0.0167 and -0.0169 respectively, which are both within 1σ of our updated values.

4.2 Unphysicality

We define the unphysicality of the density matrix, $\mathcal{U}$, as the sum of the absolute values of its negative eigenvalues. The precise origin of this unphysicality is not yet fully understood, although several contributing factors are likely. First, a small degree of unphysicality is expected due to the nature of our MC simulations: statistical fluctuations in the data, along with the finite number of events used to compute the expectation values, could lead to deviations. We can estimate the uncertainty on each of the eigenvalues of the density matrix so as to establish whether the unphysicality of each eigenvalue is due to statistical fluctuations or not. We use the MC sampling method outlined in Section 3.4, to generate a distribution of 10,000 density matrices, and then calculate the standard deviation of each eigenvalue of the density matrix. Using this method, we quantify the uncertainty on each eigenvalue of the density matrix to be ~ 0.005 for the W^+W^- system, and ~ 0.02 for the ZZ system. This means that we can expect negative eigenvalues up to ~ -0.01 for the W^+W^- system, and ~ -0.04 for the ZZ system (values within 2σ are to be expected). However, although some regions of phase

	ZZ
$\alpha^{(1)}_{1,0}$	0.00013(30)
$\alpha^{(1)}_{2,0}$	0.0299(3)
$\alpha^{(3)}_{1,0}$	0.00032(30)
$\alpha^{(3)}_{2,0}$	0.0305(3)
$\alpha^{(1)}_{2,-2}$	-0.0167(2)
$\alpha^{(3)}_{2,-2}$	-0.0171(2)
$\gamma_{1,0,1,0}$	-0.00173(9)
$\gamma_{2,0,2,0}$	0.00189(9)

Table 1: Angular coefficients for the ZZ production process at LO and $\sqrt{s} = 13$ TeV. Using the inclusive setup previously defined in Section 3.1, and a fixed renormalisation and factorisation scale $\mu_R = \mu_F = \mathrm{M}_Z$. Numerical uncertainties are shown in parentheses for the final digit(s). Adapted from [17].



(a) W^+W^- (b) ZZ

Figure 2: Unphysicality heatmaps for W^+W^- and ZZ events. The heatmaps show the distribution of unphysicality values across the phase space, with a label in each region indicating the value of unphysicality. It can be seen that the W^+W^- system seems to have a more structured distribution of unphysicality values, while the ZZ system is slightly more uniform in its unphysicalities. In addition, the W^+W^- system has a much larger range of unphysicality values, with extreme regions having unphysicalities of ~ 0.6, while the ZZ system has a maximum unphysicality of ~ 0.3.

space can be expected to have unphysicalities of this magnitude, this uncertainty does not explain the regions with higher deviations, as can be seen in Figure 2.

It is interesting to note the difference in the unphysicality pattern of the two systems. The W^+W^- system displays a wider range and more structure in its distribution: regions with lower $\cos\Theta$ (transverse scattering) seem to exhibit small unphysicalities that are likely to be consistent with statistical fluctuations, whilst regions with higher $\cos\Theta$ show larger unphysicalities. Meanwhile, the ZZ system shows a narrower range and less structure overall, which could be related to the larger amount of eigenvalue uncertainty compared to the W^+W^- system. Although slightly more events were generated for the W^+W^- sample (200,000 per region compared to 150,000 for the ZZ system), the difference is insufficient to account for this discrepancy. This comparison is interesting, as it suggests that the differences in the physics of the two systems plays a role in how the unphysicalities are distributed, and how sensitive they are to the reconstruction of the density matrix.

A second possible source of unphysicality could be due to the assumptions in the method of reconstruction of the density matrix itself. Specifically, it assumes that the full angular distributions of the leptons are known in order to calculate the expectation values. However, when reconstructing the density matrix for each region of phase space, the validity of this assumption becomes questionable. This issue is related to the size of the regions; if the regions are too small, then the angular distributions of the leptons in that region may not be well enough defined, and unphysicalities may arise.

Finally, it is important to recall that the density matrix is intended to describe the full ensemble of quantum states accessible to the system. Therefore, if we restrict the density matrix to a certain region of the phase space, then we are indirectly excluding part of this ensemble. To investigate this effect, additional tests were performed in which multiple regions of phase space were combined before reconstructing the density matrix. In these cases, the reconstructed density matrices tended to have eigenvalues that were less negative, indicating that the overall state was closer to being physical. For example, in the W^+W^- system, when combining regions with $200 < \mathrm{M}_{WW} < 600\,\mathrm{GeV}$, and $0 < \cos\Theta < 0.4$, the unphysicality was reduced to ~ 0.04, compared to the average unphysicality of ~ 0.07 for the individual regions. Furthermore, when combining an even larger number of regions, namely from $200 < \mathrm{M}_{WW} < 400\,\mathrm{GeV}$ and $0 < \cos\Theta < 0.9$, the unphysicality for the whole region was reduced to ~ 0.13, compared to the average unphysicality of ~ 0.18 for the individual regions. This is not a very significant reduction, but it does indicate that the unphysicality is reduced when the density matrix is reconstructed for a larger region of phase space. Therefore, it is likely that the unphysicality is due to a combination of different factors.

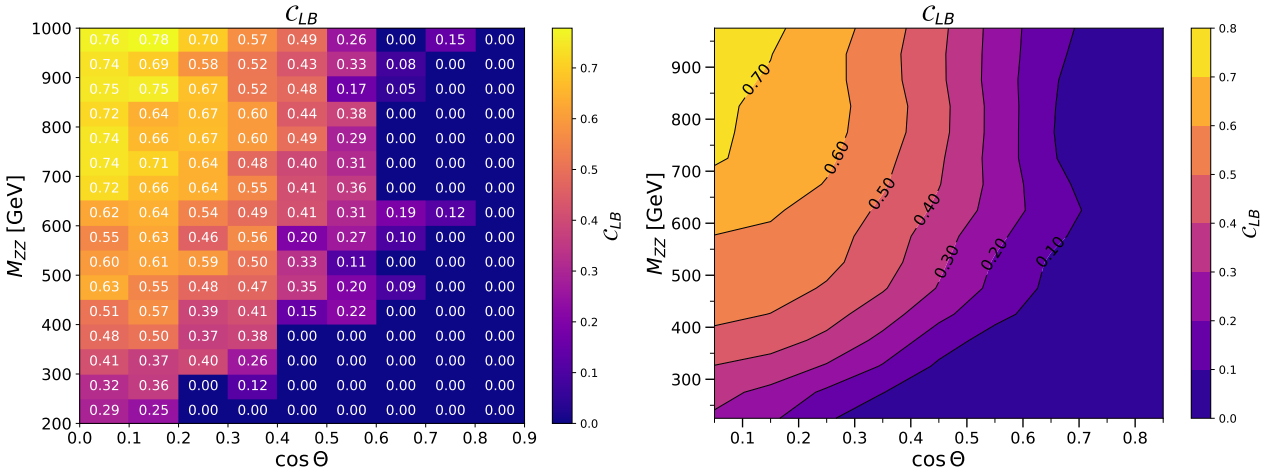
Despite these limitations, the work done in [25, 26] computes the density matrix analytically for each event individually. This avoids reconstruction of the density matrix from the angular distributions of the leptons, and instead expresses it directly as a function of the kinematic variables that define the phase space. This method is a sound theoretical approach, and is not expected to encounter any of the complications listed above. However, in terms of practical implementation, the reconstruction of the density matrix from data is essential for testing the validity of these theoretical predictions.

4.3 $pp \to ZZ$

Here, we present our results for the $pp \to ZZ$ quantum observables. We start by discussing the results for the concurrence, and then we present the results for the Bell operator. All of our results are presented in the same way, with a heatmap and contour plot for each observable. The contour plots have had a Gaussian Filter [44] applied, in order to smooth out the data and make it easier to interpret. The heatmaps, however, show the raw data, with no smoothing applied.

4.3.1 Concurrence

Below, we present the results for the lower bound of the concurrence, $\mathcal{C}_{LB}$, for the ZZ system.



(a) Heatmap **(b)** Contour plot

Figure 3: Concurrence lower bound heatmap and contour plot for $pp \to ZZ$ events. The heatmap shows the distribution of concurrence values across the phase space, while the contour plot provides a clearer interpretation of the distribution of concurrence. The colour scale indicates the level of concurrence, with darker regions representing lower values.

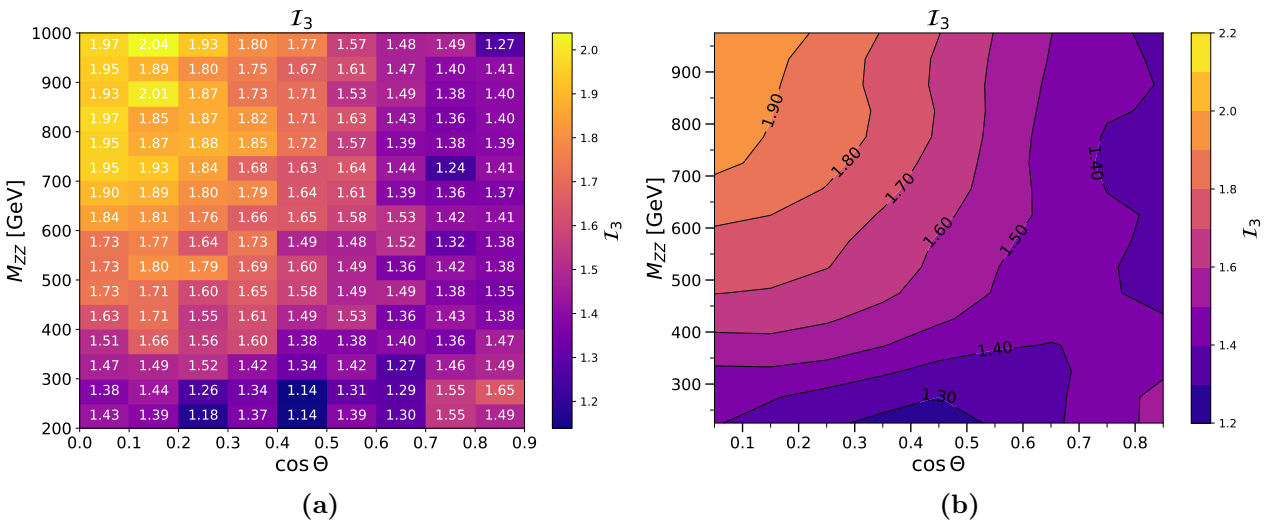
We can see that the concurrence is mostly zero in regions where $\cos\Theta$ is greater than 0.6, whereas the events with lower $\cos\Theta$ have a much higher concurrence. These values are similar to the results shown in [25, 26], where the concurrence can be seen to be greater than 0.5 in the regions with $\mathrm{M}_{ZZ} > 400\,\mathrm{GeV}$ and $\cos\Theta < 0.4$. It should be noted that some deviation from the literature is expected, as we employ a fundamentally different approach to calculating quantum observables—reconstructing the density matrix directly from MC data. Furthermore, since we are clipping negative eigenvalues, this could also lead to some differences in the results, as we are essentially destroying some of the original information in the density matrix, in favour of physical validity.

Furthermore, due to the limited amount of time available for this project, a full error propagation analysis for the uncertainty on the lower bound of the concurrence shown in Figure 3 was not performed. Instead, priority was given to quantifying the uncertainty on the eigenvalues of the reconstructed density matrix and the Bell operator expectation value, since these were more directly relevant to the validation and interpretation of our results.

4.3.2 Bell Operator

Now, we discuss the results for the Bell operator, or the violation of the Bell inequality, $\mathcal{I}_3$, for the ZZ system. Below, we present the heatmap and contour plot for the expectation value of the optimised Bell operator, $\langle \mathcal{O}_B \rangle_{max}$, which is defined in Equation 2.3.10.

Before analysing the results, it is important to note that, for the ZZ system, the uncertainty in each region of the phase space ranged from approximately 0.08 to 0.13; the higher uncertainties occurred in regions with greater $\cos\Theta$ values. In addition, a systematic uncertainty due to the eigenvalue clipping



(a) **(b)**

Figure 4: Bell operator heatmap and contour plot for $pp \to ZZ$ events. The heatmap shows the distribution of Bell operator values across the phase space, while the contour plot provides an overall view of the structure of the Bell operator. The colour scale indicates the level of Bell operator values, with darker regions representing lower values.

has not been applied to the error propagation procedure, as it is difficult to quantify.

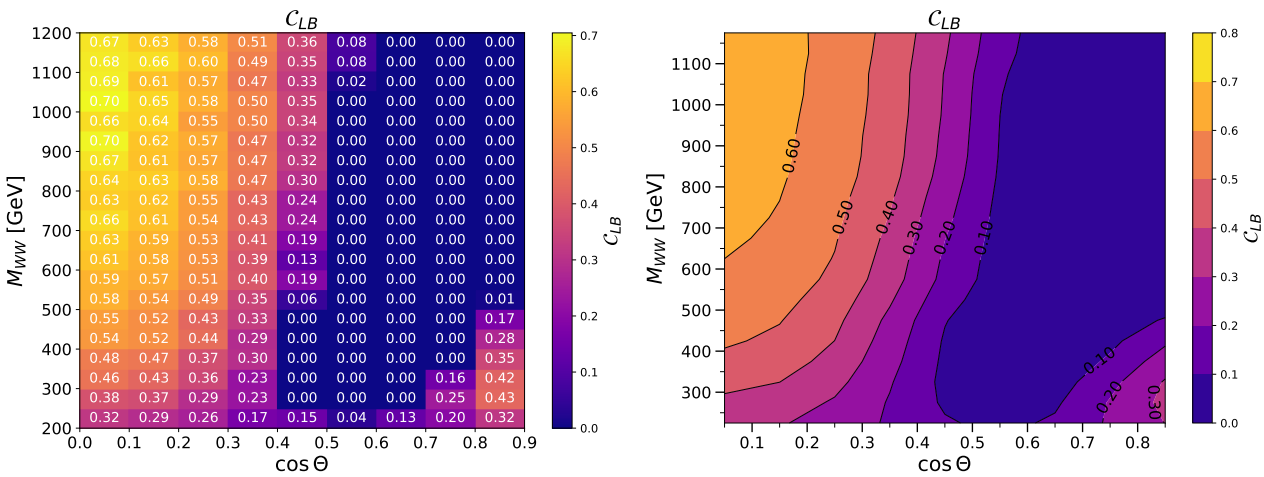
The results for the Bell operator (Figure 4) show a similar pattern to the concurrence, with the regions of higher $\cos\Theta$ having lower values of the Bell operator. When considering which regions of the phase space show signs of Bell inequality violation, we can see in Figure 4a that there are only two regions with a Bell operator value that exceeds 2. However, taking into account the uncertainties, we cannot confirm or deny whether these regions are indeed violating the Bell inequality, since many regions are within 1σ of the value of 2. According to the studies shown in [25, 26] for the ZZ system, the regions that violate the Bell inequality are those with $\mathrm{M}_{ZZ} > 500\,\mathrm{GeV}$ and $\cos\Theta < 0.3$, meaning that our results are within approximately 2σ of the literature. It should be noted that the number of events generated for the ZZ system was 150,000 per region, and one may question whether this is sufficient to obtain a good estimate of the Bell operator. Initially, only 50,000 events per region were generated, but this was later increased threefold in an effort to reduce statistical uncertainties. While further increasing the number of events would likely have improved the precision of the final results, this was ultimately limited by time constraints—both in terms of computational resources and the overall timeline of the project. Since the final uncertainty calculations were performed relatively late in the analysis, it became apparent only in hindsight that a larger dataset might have been beneficial. However, given that each dataset required approximately one week to generate, further increases in statistics were not feasible within the available time.

It is also worth noting that the study shown in [28] aimed to reconstruct the density matrix from angular distributions for both the W^+W^- and ZZ systems, however they did not present any explicit results for the Bell operator. Instead, they mentioned that they had calculated the expectation value of the Bell operator, but that they could not conclude whether they had observed any violation of the Bell inequality, due to the uncertainties in their simulation.

4.4 $pp \to W^+W^-$

Here, we present our results for the W^+W^- quantum observables in the same manner as for the ZZ system.

4.4.1 Concurrence



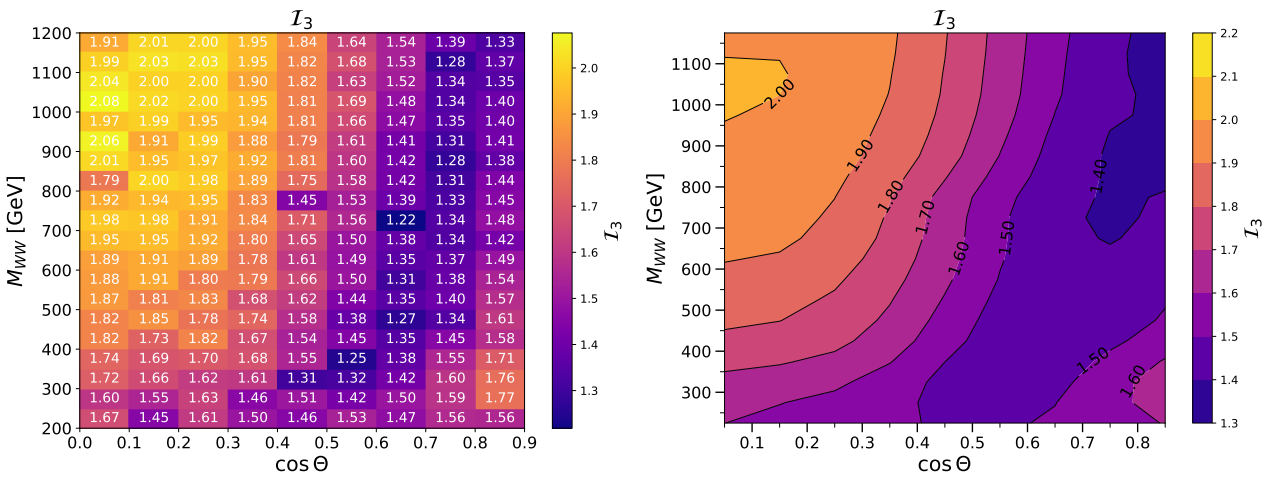
(a) **(b)**

Figure 5: Concurrence heatmap and contour plot for W^+W^- events. The heatmap displays the distribution of concurrence values across the phase space, while the contour plot provides a clearer depiction of regions with high and low concurrence. The colour scale indicates the level of concurrence, with darker regions representing lower values.

For the W^+W^- system, we can see in Figure 5 a similar structure to the ZZ system, with the regions of higher $\cos\Theta$ having lower values of the concurrence, excecpt for one key distinction, in the low M_{WW} – low $\cos\Theta$ region, where the concurrence seems to increase again. This type of 'tail' structure in the concurrence is not present in our ZZ system, but can be seen in the results shown in [25, 26]. Another key difference is that the W^+W^- system has a lower maximum value of the concurrence, with the maximum regions shown in the contour plot having a value of > 0.6, while the maximum value for the ZZ system was > 0.7. This is also in direct agreement with the results shown in [25, 26], where the maximum regions for the W^+W^- system were those with $\mathrm{M}_{WW} > 600\,\mathrm{GeV}$ and $\cos\Theta < 0.2$.

4.4.2 Bell Operator

In this final section, we present the results for the Bell operator expectation value for the W^+W^- system. It is interesting to note that the uncertainty on the Bell operator expectation value for the W^+W^- system was much lower than that of the ZZ system, with the uncertainty in each region of the phase space ranging from ~ 0.02 to ~ 0.03. This is perhaps surprising, given that the statistics of the W^+W^- system were not that much larger than those of the ZZ system, which indicates that there must be some underlying structure in the physics of these two systems that is affecting the sensitivity of the Bell operator to the statistics. Below, in Figure 6, we present the heatmap and contour plot for the Bell operator expectation value, or Bell inequality, $\mathcal{I}_3$, for the W^+W^- system.



(a) **(b)**

Figure 6: Bell operator heatmap and contour plot for W^+W^- events. The heatmap shows the distribution of Bell operator values across the phase space, while the contour plot provides a clearer representation of the regions of high and low Bell operator values. The colour scale indicates the level of Bell operator values, with darker regions representing lower values.

The Bell operator for the W^+W^- system exhibits a structure similar to that of its concurrence plots, and its expectation values are generally higher than those observed in the ZZ system. In several regions, Bell operator values exceed the classical bound of 2. However, due to statistical uncertainties, we cannot indisputably claim a violation of the Bell inequality. Many of these values remain within 1σ of the classical bound, despite the uncertainties being smaller than those in the ZZ system.

When comparing our results to those reported in [25, 26], we observe that our Bell operator values are consistently lower. This raises the possibility that our method of eigenvalue clipping may be suppressing the observed quantum correlations. Specifically, our current approach enforces strict positive semi-definiteness by hard-clipping negative eigenvalues of the reconstructed density matrix. While this ensures physical validity, it may also remove meaningful information, as we don't know how sensitive the system is to this eigenvalue clipping method.

To explore alternatives, we briefly investigated a *soft-lift* procedure, in which highly negative eigenvalues would be raised closer to zero, while smaller eigenvalues would be minimally affected. Although such a method may preserve more of the original structure of the data, we ultimately chose to present results based on strictly physical (positive semi-definite) matrices, as this approach is more clearly motivated by the requirements of quantum mechanics.

Another interesting observation related to the Bell operator of the W^+W^- system is that in [25], it can be seen in Figure 11 that the region with $0.8 < \cos\Theta < 1$ and $300 < \mathrm{M}_{WW} < 500\,\mathrm{GeV}$ shows a significant deviation from the rest of the phase space. This deviation, however, is not present in the work conducted by [26] (Figure 12) and in our own results. What is perhaps even more interesting about this region is that it also so happens to be the region where we observed the highest amount of unphysicality for the W^+W^- system. This could be a coincidence, but it is worth noting, as it could be a sign of some underlying physical effect that is causing this discrepancy.

5 Conclusion and Outlook

In this project, we have successfully reconstructed the density matrix for the W^+W^- and ZZ systems using the angular distributions of the decay products from Monte Carlo simulations. This allowed us to compute key quantum observables relevant to the study of entanglement, with the goal of quantifying quantum correlations between pairs of electroweak bosons.

As part of this work, we also presented an erratum to [17], correcting the angular coefficients for the ZZ system and demonstrating good agreement with results previously reported in the literature. We then evaluated the concurrence lower bound and the Bell operator expectation value for both systems, discussing the differences in their behaviour and physical implications. The role of statistical and systematic uncertainties was carefully considered, particularly regarding their impact on our ability to robustly claim a violation of Bell's inequality. These uncertainties also helped interpret instances of unphysical behaviour, such as negative eigenvalues in the reconstructed density matrices. We proposed and applied a method to project the density matrix to a physical (positive semi-definite) state, ensuring valid quantum observables, while also discussing the trade-offs involved in this procedure. A comparison with existing literature highlighted discrepancies in the magnitude of quantum observables, which we attributed to methodological differences, including our use of full numerical density matrix reconstruction and eigenvalue clipping, as opposed to purely analytical approaches.

Future work could focus on a more detailed analysis of the eigenvalue clipping method and its influence on the calculated observables. In particular, the *soft-lift* method discussed in Section 4.4.2 may offer a more balanced approach by preserving small fluctuations while mitigating large unphysical contributions. Alternatively, a method which only clips negative eigenvalues beyond 5σ or some multiple of the eigenvalues uncertainty could also be tested. Additionally, generating larger datasets would reduce statistical noise, potentially leading to more reliable quantum state reconstructions. Incorporating higher-order corrections—such as QCD or EW loop effects—during event generation in `MadGraph5` could also improve the physical realism of the simulation, providing more accurate input for density matrix construction. Lastly, extending this framework to experimental data would enable direct comparisons between theoretical predictions and real-world collider observations.

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References

[1] A. Einstein, B. Podolsky and N. Rosen, *Can quantum mechanical description of physical reality be considered complete?*, *Phys. Rev.* **47** (1935) 777.

[2] J.S. Bell, *On the Einstein-Podolsky-Rosen paradox*, *Physics Physique Fizika* **1** (1964) 195.

[3] J.F. Clauser, M.A. Horne, A. Shimony and R.A. Holt, *Proposed experiment to test local hidden variable theories*, *Phys. Rev. Lett.* **23** (1969) 880.

[4] S.J. Freedman and J.F. Clauser, *Experimental Test of Local Hidden-Variable Theories*, *Phys. Rev. Lett.* **28** (1972) 938.

[5] G. Weihs, T. Jennewein, C. Simon, H. Weinfurter and A. Zeilinger, *Violation of Bell's inequality under strict Einstein locality conditions*, *Phys. Rev. Lett.* **81** (1998) 5039 [quant-ph/9810080].

[6] D. Bouwmeester, J.-W. Pan, K. Mattle, M. Eibl, H. Weinfurter and A. Zeilinger, *Experimental quantum teleportation*, *Nature* **390** (1997) 575.

[7] D. Collins, N. Gisin, N. Linden, S. Massar and S. Popescu, *Bell Inequalities for Arbitrarily High-Dimensional Systems*, *Phys. Rev. Lett.* **88** (2002) 040404.

[8] M.M. Wilde, *Quantum Information Theory*, Cambridge University Press (4, 2013), 10.1017/cbo9781139525343.

[9] M.A. Nielsen and I.L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (6, 2012), 10.1017/cbo9780511976667.

[10] N. Gisin, G. Ribordy, W. Tittel and H. Zbinden, *Quantum cryptography*, *Reviews of Modern Physics* **74** (2002) 145–195.

[11] V. Dunjko and H.J. Briegel, *Machine learning & artificial intelligence in the quantum domain: a review of recent progress*, *Reports on Progress in Physics* **81** (2018) 074001.

[12] Y. Afik et al., *Quantum Information meets High-Energy Physics: Input to the update of the European Strategy for Particle Physics*, 2504.00086.

[13] ATLAS collaboration, *Measurement of $W^{\pm}Z$ production cross sections and gauge boson polarisation in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *Eur. Phys. J. C* **79** (2019) 535 [1902.05759].

[14] ATLAS collaboration, *Observation of gauge boson joint-polarisation states in $W^{\pm}Z$ production from pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, *Phys. Lett. B* **843** (2023) 137895 [2211.09435].

[15] Particle Data Group collaboration, *Review of Particle Physics*, *Phys. Rev. D* **98** (2018) 030001.

[16] H. El Faham, G. Pelliccioli and E. Vryonidou, *Triple-gauge couplings in LHC diboson production: a SMEFT view from every angle*, *JHEP* **08** (2024) 087 [2405.19083].

[17] F. Tcach, "The LHC as a Quantum Experiment: Quantum Entanglement with W boson pairs." Dec., 2024.

[18] R. Shankar, *Principles of Quantum Mechanics*, Springer, New York, 2nd ed. (1994), 10.1007/978-1-4757-0576-8.

[19] W.K. Wootters, *Entanglement of formation of an arbitrary state of two qubits*, *Physical Review Letters* **80** (1998) 2245–2248.

[20] R. Horodecki, P. Horodecki, M. Horodecki and K. Horodecki, *Quantum entanglement*, *Rev. Mod. Phys.* **81** (2009) 865 [quant-ph/0702225].

[21] P. Rungta, V. Bužek, C.M. Caves, M. Hillery and G.J. Milburn, *Universal state inversion and concurrence in arbitrary dimensions*, *Physical Review A* **64** (2001) .

[22] F. Mintert and A. Buchleitner, *Observable entanglement measure for mixed quantum states*, *Phys. Rev. Lett.* **98** (2007) 140505.

[23] C.-J. Zhang, Y.-X. Gong, Y.-S. Zhang and G.-C. Guo, *Observable estimation of entanglement for arbitrary finite-dimensional mixed states*, *Physical Review A* **78** (2008) .

[24] M. Nakahara and T. Ohmi, *Quantum Computing: From Linear Algebra to Physical Realizations*, CRC Press, Boca Raton, FL (2008), 10.1201/9781420012293.

[25] M. Fabbrichesi, R. Floreanini, E. Gabrielli and L. Marzola, *Bell inequalities and quantum entanglement in weak gauge boson production at the LHC and future colliders*, *Eur. Phys. J. C* **83** (2023) 823 [2302.00683].

[26] R. Aoude, E. Madge, F. Maltoni and L. Mantani, *Probing new physics through entanglement in diboson production*, *JHEP* **12** (2023) 017 [2307.09675].

[27] J.A. Aguilar-Saavedra, A. Bernal, J.A. Casas and J.M. Moreno, *Testing entanglement and Bell inequalities in H→ZZ*, *Phys. Rev. D* **107** (2023) 016012 [2209.13441].

[28] R. Ashby-Pickering, A.J. Barr and A. Wierzchucka, *Quantum state tomography, entanglement detection and Bell violation prospects in weak decays of massive particles*, *JHEP* **05** (2023) 020 [2209.13990].

[29] R. Rahaman and R.K. Singh, *Breaking down the entire spectrum of spin correlations of a pair of particles involving fermions and gauge bosons*, *Nucl. Phys. B* **984** (2022) 115984 [2109.09345].

[30] J.A. Aguilar-Saavedra, A. Bernal, J.A. Casas and J.M. Moreno, *Testing entanglement and Bell inequalities in H→ZZ*, *Phys. Rev. D* **107** (2023) 016012 [2209.13441].

[31] A.J. Barr, M. Fabbrichesi, R. Floreanini, E. Gabrielli and L. Marzola, *Quantum entanglement and Bell inequality violation at colliders*, *Prog. Part. Nucl. Phys.* **139** (2024) 104134 [2402.07972].

[32] M. Grossi, G. Pelliccioli and A. Vicini, *From angular coefficients to quantum observables: a phenomenological appraisal in di-boson systems*, 2409.16731.

[33] G.B. Arfken, H.J. Weber and F.E. Harris, *Mathematical Methods for Physicists*, Academic Press, Boston, 7th ed. (2013), 10.1016/B978-0-12-384654-9.00015-3.

[34] M. Gell-Mann, *Symmetries of baryons and mesons*, *Phys. Rev.* **125** (1962) 1067.

[35] J. Alwall, R. Frederix, S. Frixione, V. Hirschi, F. Maltoni, O. Mattelaer et al., *The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations*, *JHEP* **07** (2014) 079 [1405.0301].

[36] A. Denner and G. Pelliccioli, *NLO EW and QCD corrections to polarized ZZ production in the four-charged-lepton channel at the LHC*, *JHEP* **10** (2021) 097 [2107.06579].

[37] D.Y. Bardin, A. Leike, T. Riemann and M. Sachwitz, *Energy Dependent Width Effects in e+ e- Annihilation Near the Z Boson Pole*, *Phys. Lett. B* **206** (1988) 539.

[38] A. Buckley, J. Ferrando, S. Lloyd, K. Nordström, B. Page, M. Rüfenacht et al., *LHAPDF6: parton density access in the LHC precision era*, *Eur. Phys. J. C* **75** (2015) 132 [1412.7420].

[39] NNPDF collaboration, *Parton distributions from high-precision collider data*, *Eur. Phys. J. C* **77** (2017) 663 [1706.00428].

[40] N.J. Higham, *Computing a nearest symmetric positive semidefinite matrix*, *Linear Algebra and its Applications* **103** (1988) 103.

[41] P. Bevington and D. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, McGraw-Hill international editions: Physics series, McGraw-Hill (2003).

[42] C.R. Harris, K.J. Millman, S.J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau et al., *Array programming with NumPy*, *Nature* **585** (2020) 357.

[43] C.E. Papadopoulos and H. Yeung, *Uncertainty estimation and monte carlo simulation method*, *Flow Measurement and Instrumentation* **12** (2001) 291.

[44] P. Virtanen, R. Gommers, T.E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau et al., *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python*, *Nature Methods* **17** (2020) 261.

Appendices

A Gell-Mann matrices

The Gell-Mann matrices are a set of eight 3×3 matrices that form a basis for the Lie algebra of the special unitary group SU(3). They are defined as follows:

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

$$\lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

B Irreducible Tensor Operator Parametrisation

The irreducible tensor operator (ITO) parametrisation is shown below, where the T^l_m are the irreducible tensor operators, where l is the rank of the tensor operato (orbital angular momentum quantum number), and m is the magnetic quantum number (azimuthal quantum number).

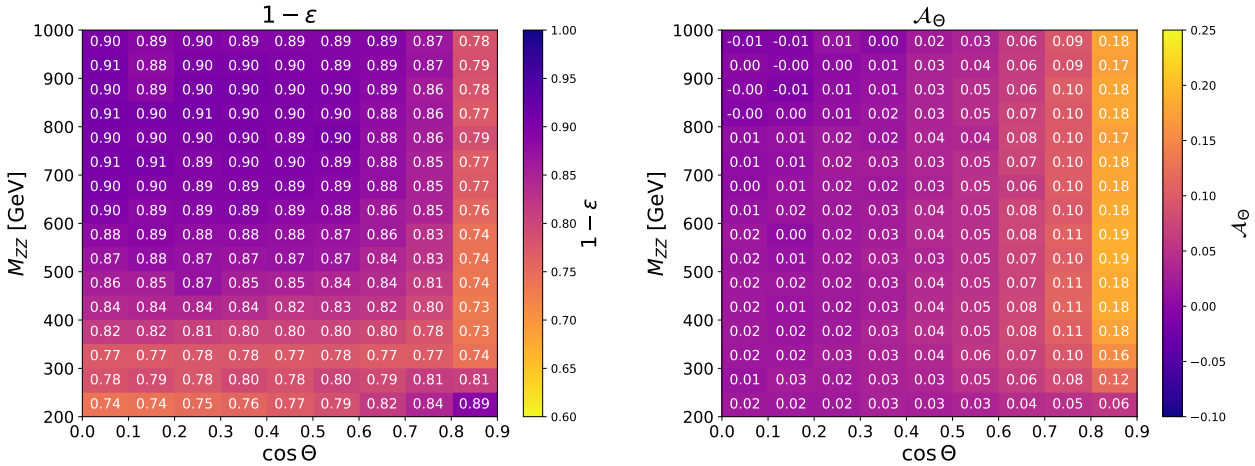
$$T^1_{-1} = \sqrt{\frac{3}{2}} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad T^1_0 = \sqrt{\frac{3}{2}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$T^1_1 = \sqrt{\frac{3}{2}} \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \qquad T^2_{-2} = \sqrt{3} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$T^2_{-1} = \sqrt{\frac{3}{2}} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \qquad T^2_0 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

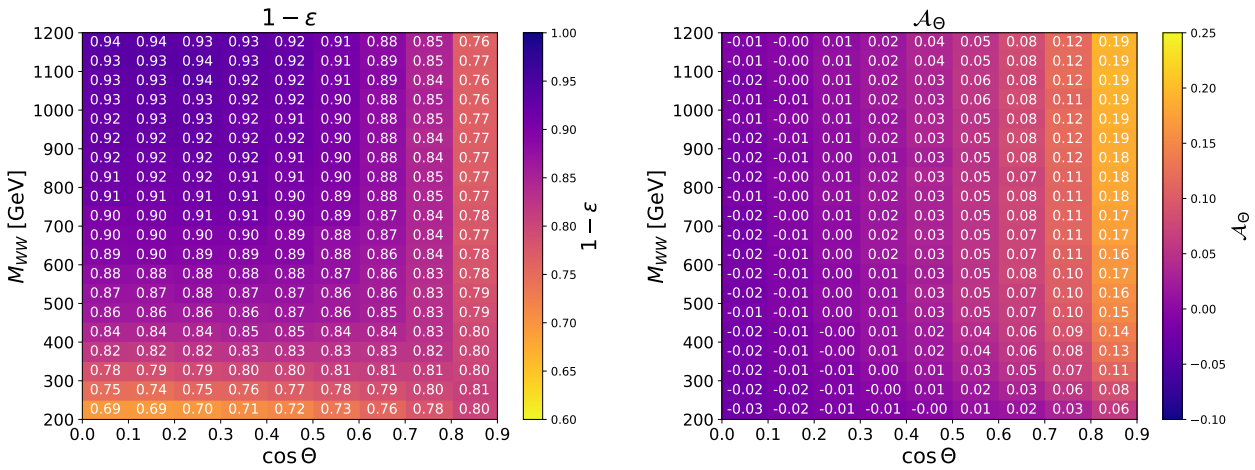
$$T^2_1 = \sqrt{\frac{3}{2}} \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \qquad T^2_2 = \sqrt{3} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

C Phase-Space Validation



(a) Uniformity **(b)** Asymmetry

Figure 7: Uniformity and $\cos\Theta$ asymmetry heatmaps for the ZZ system.



(a) Uniformity **(b)** Asymmetry

Figure 8: Uniformity and $\cos\Theta$ asymmetry heatmaps for the W^+W^- system.